

GRADE 3 CAPSTONE PROJECT GUIDE

Design Your Own Circuit!

The Grand Capstone Engineering Challenge

Welcome, Junior Circuit Inventor! Think about your favorite gadget at home. Is it a flashlight? A toy robot? Or a musical doorbell? All of those amazing toys have one big thing in common: inside them, hidden from view, is a working circuit. A circuit is simply a complete, circular path that lets electric power travel around to complete chores for us. Today, you get to use your own imagination to sketch, plan, and build a unique gadget from scratch!

Our Core Building Blocks

Building Block 1: The Power Source (Battery)

Every single electronic machine needs a starting push. We use a Battery as our ultimate power pump! Think of a battery like a friendly heart that pumps electric current out into the circuit line.

- **The Plus Side (+):** The high-energy exit gate. Electricity comes rushing out from here, excited to start its journey.
- **The Minus Side (-):** The return gate. After electricity travels through your outputs, it circles back here to rest and recharge.

Building Block 2: The Action Outputs (LED & Buzzer)

In engineering, components that do actions are called Outputs. You have two amazing choices today:

- **The Bright LED Light:** Turns electricity into a beautiful glow. Perfect for a lighthouse beam, toy headlight, or secret signal lamp!
- **The Loud Piezo Buzzer:** Turns electric energy into a buzzing vibration sound! Perfect for a morning alarm clock, game-show button, or safety siren.

Building Block 3: The Path Controllers (Switches)

A switch acts like a bridge operator over a river to control whether our action outputs are running or resting:

- **Bridge Closed (ON):** When you press the switch down, the metal parts touch, completing the path so electricity can fly across!
- **Bridge Open (OFF):** When you let go or flip it up, a gap is made in the wire, stopping electricity instantly in its tracks.

Material Pre-Check Sheet

Component Check	What to Look For
Battery Snap	Are the red and black insulation coatings clean without any tears or exposed interior metal?
LED Wire Legs	Is the longer positive leg easily visible? Make sure it isn't bent or broken.
Buzzer Leads	Check the red (+) wire and black (-) wire coming out of the buzzer casing for clean metal tips.
Clip Wires	Do the metal alligator clips pinch tightly onto the component leads without slipping off?

Step 1: Draw Your Secret Blueprint

Real engineers never build blindly without a map. Before you touch a single real wire, you must sketch a blueprint layout on your project sheet! Make sure your pencil line forms a complete, unbroken loop from the battery, through the switch, into your chosen output, and back home to the negative terminal.

Step 2: Wire It Together Step-by-Step

Time to assemble! Choose the step-by-step track below that matches the creative gadget you are building today:

Option A: The LED Light Circuit Track

1. Clip a jumper wire from the **Positive Red terminal** of your battery holder over to the incoming terminal of your switch.
2. Connect a second wire from the outgoing terminal of your switch to the **Long Leg (+)** of your LED light bulb.
3. Connect a final wire from the **Short Leg (-)** of your LED light directly back to the **Negative Black terminal** of your battery pack.

Option B: The Piezo Buzzer Circuit Track (New!)

1. Clip a jumper wire from the **Positive Red terminal** of your battery holder over to the incoming terminal of your switch.
2. Connect a second wire from the outgoing terminal of your switch directly to the **Positive Red wire (+)** of your Piezo Buzzer.
3. Connect a final wire from the **Negative Black wire (-)** of your Piezo Buzzer directly back to the **Negative Black terminal** of your battery pack.

Troubleshooting & Engineering Detective Work

What happens if you click your switch to ON, but your gadget stays completely dark or quiet? Don't worry! That is exactly what real engineers call Troubleshooting. Check these two common bugs:

- **The Backwards Bug:** For LEDs, make sure the long leg points to the red wire side. For Buzzers, ensure the red buzzer wire is connected to the switch output, not the negative rail!
- **The Loose Clip Leak:** Wiggle your alligator clips gently. Make sure they are biting down hard onto bare metal wire, not the plastic protective coating.

CRITICAL SAFETY CHECK: The Heat Test

If your battery pack or any wire feels warm or hot to the touch at any point, disconnect it immediately! This means a wire is bypassing your output and causing a short circuit. Ask an adult helper to inspect the lines before restarting your experiment.

Capstone Finale: Present Your Invention!

Give your device an awesome name, stand proud at your lab bench, and explain your new gadget to the class:

*"I built a circuit named _____ that uses a switch to power up a _____ whenever
a user presses the bridge down!"*